

Probability Review

COMPSCI 753

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Outline

- Linearity of expectation
- Warm-up exercises

$$E[Y_1 \star Y_0] = E[Y_1] \star E[Y_0]$$

$\star = + / -$

- **[Book]** Algorithm Design by Jon Kleinberg and Éva Tardos -- *Chapter 13*



Finite Probability Space

- Simple probability statements
 - S1: “If a fair coin is flipped, the probability of heads is $1/2$.”
 - S2: “If a fair dice is rolled, the probability of 6 is $1/6$.”
- Finite probability space: all possible outcomes
 - S1: $\Omega = \{\text{'heads'}, \text{'tails'}\}$
 - S2: $\Omega = \{1, 2, 3, 4, 5, 6\}$
- Compute the probability of a particular outcome
 - $\Pr[\text{coin} = \text{'heads'}] = 1/2$
 - $\Pr[\text{dice} = 6] = 1/6$

Definition: Event

- Event
 - A specific event E is defined by a **subset of outcomes** from the finite probability space
 - We are interested in $\Pr[E]$, i.e. the probability that the event E occurs
 - Complement event $\bar{E}$: the event E does not occur, i.e. $\Pr[\bar{E}] = 1 - \Pr[E]$
- Example
 - Rolling two fair dices, what is the probability that the event of “2 dices are the same” occurs?
 - $\Omega = \{(1,1), \dots, (1,6), \dots, (6,1), \dots, (6,6)\}$
 - $\Pr[2 \text{ dices are the same}] = \Pr[(1,1)] + \Pr[(2,2)] + \dots + \Pr[(6,6)]$
 - $\Pr[2 \text{ dices are different}] = 1 - \Pr[2 \text{ dices are the same}]$

Independent Events

- Two events A and B are independent if their outcomes do not affect each other
 - A: The event that flipping the 1st fair coin returns 'heads'
 - B: The event that flipping the 2nd fair coin returns 'heads'
- $\Pr[A \wedge B] = \Pr[A] \cdot \Pr[B]$
 - $\Pr[1^{st} \text{ coin} = \text{'heads'} \wedge 2^{nd} \text{ coin} = \text{'heads'}] = 1/2 \times 1/2 = 1/4$
- If $E_1, \dots, E_n$ are independent events, then $\Pr[E_1 \wedge \dots \wedge E_n] = \Pr[E_1] \dots \Pr[E_n]$

E.g., Monte Carlo Algorithm

- Problem: Given an array of n elements $A[0, \dots, n-1]$ where n is even. Half of the elements are 0s, the other half are 1s.
- Goal: Find an index containing 1

Monte Carlo Solution:

```
1. Repeat 100 times:  
2.     k = randint(n)  
3.     If A[k] == 1  
4.         Return k  
5. Return "Failed"
```

- Probability of getting 0 in the 1st times: $\frac{1}{2}$
- Probability of getting 0s 100 times: $\left(\frac{1}{2}\right)^{100}$

Random Variables

- Informally, X is a random variable if
 - X is an outcome of a random process
 - X presents a value which we are uncertain of
- Examples
 - Random process: rolling a fair dice, define X as the value of the dice
 - $X = 1/2/3/4/5/6$ with the same probability $1/6$ (i.e., `randint(6)`)
 - Random process: flipping a fair coin, if 'heads', we return 1; otherwise, we return 0
 - X : the random variable of this process outcome
 - $X = 0/1$ with the same probability $1/2$ (i.e., `Bernoulli(0.5)`)

Expectation of a Random Variable

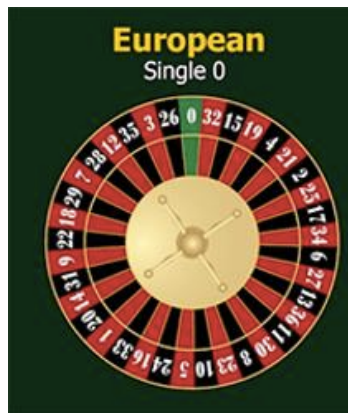
- Expectation: Given a discrete random variable X with value in $\{0, 1, \dots, \infty\}$, its expectation $E[X]$ (e.g., the average value of X) is defined by:

$$E[X] = \sum_{j=0}^{\infty} j \times \Pr[X = j]$$

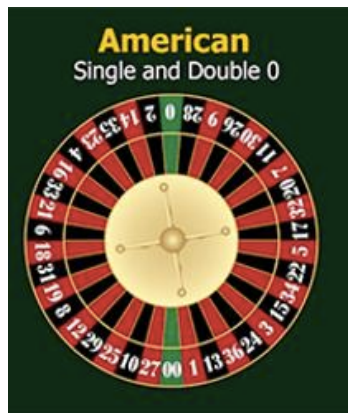
- Example:
 - Rolling a fair dice, expectation of the dice value is:
 $1 \times 1/6 + 2 \times 1/6 + 3 \times 1/6 + 4 \times 1/6 + 5 \times 1/6 + 6 \times 1/6 = 7/2$
 - Given a random variable X generated by **Bernoulli(0.5)**:
 - $\Pr[X = 1] = 0.5$ and $\Pr[X = 0] = 0.5$
 - $E[X] = 1 \times 0.5 + 0 \times 0.5 = 0.5$



E.g., Roulette Game in Casino



or



- Rule
 - If I bet on Reds and the ball lands on red, I win an amount equal to my bet
 - Otherwise, I lose my bet
- Questions
 - Q1: Given that I bet \$1B on Reds, what is the expectation of my outcome?
 - Q2: Which game do I have a better chance to win?

European	X	\$2B	0
	P(X)	$\frac{18}{37}$	$\frac{19}{37}$

$$E[X] = 2B \times \frac{18}{37} + 0 \times \frac{19}{37} = \frac{36}{37}B < \$1B$$

~ 0.97B

American	X	\$2B	0
	P(X)	$\frac{18}{38}$	$\frac{20}{38}$

$$E[X] = 2B \times \frac{18}{38} + 0 \times \frac{20}{38} = \frac{36}{38}B < \$1B$$

~ 0.94B

Linearity of Expectation

- If X is a 0/1 random variable, then $E[X] = \Pr[X = 1]$
 - $E[X] = 1 \times \Pr[X = 1] + 0 \times \Pr[X = 0] = \Pr[X = 1]$
- Given two random variables X and Y defined over the same finite probability space, we have $E[X + Y] = E[X] + E[Y]$
- Given n random variables $X_1, \dots, X_n$ defined over the same finite probability space, we have:
 $E[X_1 + \dots + X_n] = E[X_1] + \dots + E[X_n]$



E.g., Guessing Cards (Memoryless)



- Game: Shuffle n different cards and turn them over one at a time. Try to guess each card without remembering the previous cards (**sampling with replacement**)
- Question: What is the expected “number of correct guesses” if we predict n times?

Solutions:

- Define $X_i = 1$ if we guess correctly at the i – th time and 0 otherwise
- We have: $E[X_i] = 1/n$
- Define $X = X_1 + \dots + X_n$ as the “number of correct guesses”
- $E[X] = E[X_1 + \dots + X_n] = E[X_1] + \dots + E[X_n]$
$$= n \times (1 \times \frac{1}{n} + 0 \times \frac{n-1}{n}) = n \times \frac{1}{n} = 1$$

	1 st	2 nd	...	n^{th}
X_i	1	1	...	1
$P(X_i)$	$\frac{1}{n}$	$\frac{1}{n}$	...	$\frac{1}{n}$

E.g., Guessing Cards (Memory)



- Game: Shuffle n different cards and turn them over one at a time. Try to guess the card with memory of the previous cards (**sampling without replacement**)
- Question: What is the expected “number of correct guesses” if we predict n times?

Solutions:

- Define $X_i = 1$ if we guess correctly at the i – th time and 0 otherwise
- We have: $E[X_i] = 1/ (n - i + 1)$
- Define $X = X_1 + \dots + X_n$ as the “number of correct guesses”
- $E[X] = E[X_1 + \dots + X_n] = E[X_1] + \dots + E[X_n]$
- $E[X] = 1/n + 1/(n - 1) + \dots + 1/1 = H(n) = O(\log n)$ ← Harmonic series » $\ln(n) + 0.577$

	1 st	2 nd	...	n^{th}
X_i	1	1	...	1
$P(X_i)$	$\frac{1}{n}$	$\frac{1}{n-1}$	...	$\frac{1}{1}$

E.g., Las Vegas Algorithm

- Problem: Given an array of n elements $A[0, \dots, n-1]$ where n is even. Half of the elements are 0s, the other half are 1s.
- Goal: Find an index containing 1.

Las Vegas solutions:

```
1. Repeat:
2.   k = randint(n)
3.   If A[k] == 1
4.     Return k
```

X	1	2	...	m
$P(X)$	$\frac{1}{2}$	$\left(\frac{1}{2}\right)^2$	...	$\left(\frac{1}{2}\right)^m$

- Expected running time is $O(1)$: 2 iterations
 - X : number of iterations
 - $E[X] = 1 \times 1/2 + 2 \times (1/2)^2 + \dots + m \times (1/2)^m$
 $= 2$ (when $m \rightarrow \infty$) ← math trick

E.g., Waiting for a First Success

- Game: Flipping a biased coin where it is heads with probability p and tails with probability $1 - p$
- Question: How many independent flips X until first heads?

Solutions:

$$\begin{aligned} E[X] &= \sum_{j=1}^{\infty} jP[X = j] = \sum_{j=1}^{\infty} j(1-p)^{j-1}p \quad \leftarrow j-1 \text{ tails, 1 head} \\ &= \frac{p}{1-p} \sum_{j=1}^{\infty} j(1-p)^j = \frac{p}{1-p} \cdot \frac{1-p}{p^2} = \frac{1}{p} \end{aligned}$$

X_j	1	2	...	n
$P(X)$	p	$(1-p)p$	...	$(1-p)^{n-1}p$

Hard Trick:

Taylor Series of $f(x) = x/(1-x)^2$ at $x = a$, where $x = 1 - p$ (see next slide)

Step-by-step ...

Apply Taylor Series of $f(x) = x/(1-x)^2$ at $x = a$

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots$$

$$= 0 + \frac{\frac{d}{dx} \left(\frac{x}{(1-x)^2} \right) (0)}{1!} x + \frac{\frac{d^2}{dx^2} \left(\frac{x}{(1-x)^2} \right) (0)}{2!} x^2 + \frac{\frac{d^3}{dx^3} \left(\frac{x}{(1-x)^2} \right) (0)}{3!} x^3 + \dots$$

← Set $a = 0$
Apply the Taylor Formula

$$= 0 + \frac{1}{1!} x + \frac{4}{2!} x^2 + \frac{18}{3!} x^3 + \frac{96}{4!} x^4 + \frac{600}{5!} x^5 + \dots$$

← Evaluate Derivatives

$$= x + 2x^2 + 3x^3 + 4x^4 + 5x^5 + \dots$$

$$= \sum_{j=1}^{\infty} jx^j = \sum_{j=1}^{\infty} j(1-p)^j$$

E.g., Coupon Collector

- Game: Each box of cereal contains a coupon. There are n different types of coupons. Assuming all boxes are equally likely to contain each coupon.
- Question: “how many boxes” before we obtain ≥ 1 coupon of each type?

Solutions:

In phase i , (1) new coupon with prob. $p = (n - i)/n$
 (2) old coupon with prob. $1 - p$

→ #boxes open to get $i + 1$ new coupon with prob. $1/p = n/(n - i)$

- Phase i = time between i and $i + 1$ distinct coupons
- Let X_i = number of boxes we open in phase i
- Let $X = X_0 + X_1 + \dots + X_{n-1}$ = number of boxes in total

i	0	1	2	...	$n-1$
$E(X_i)$	1	$\frac{n}{n-1}$	$\frac{n}{n-2}$	...	$\frac{n}{n-(n-1)}$

$$E[X] = \sum_{i=0}^{n-1} E[X_i] = \sum_{i=0}^{n-1} \frac{n}{n-i} = n \sum_{i=0}^{n-1} \frac{1}{i+1} = nH(n)$$

↑
 prob. of success = $(n - i)/n \rightarrow$ expected #boxes open = $n/(n - i)$